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Exam questions

Comprehensive Explanation of Clustering and the k-means Algorithm

Question

Explain the clustering problem and describe the k-means clustering algorithm. In particular, discuss the function it minimizes and its relationship with a Gaussian model.

The Fundamental Problem of Clustering

The Search for Structure in Chaos

Imagine you have a bag of candies of various shapes and colors, randomly mixed. Your brain naturally does this: it **groups** similar candies. Round ones with round ones, square ones with square ones, red ones with red ones.

Clustering is exactly this capability, but for data. It's an **unsupervised** method - no one tells you how to group, you must discover the structure by yourself.

Formal Mathematical Definition

Given a dataset $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ with $\mathbf{x}_i \in \mathbb{R}^D$, clustering seeks a partition of $\mathcal{X}$ into K subsets $C_1, C_2, \dots, C_K$ such that:

1. **Completeness:** $\bigcup_{k=1}^K C_k = \mathcal{X}$
 2. **Exclusivity:** $C_i \cap C_j = \emptyset$ for $i \neq j$ (for hard clustering)
 3. **Cohesion:** Points in the same cluster are similar
 4. **Separation:** Points from different clusters are dissimilar
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The k-means Algorithm: A Brilliant Idea in its Simplicity

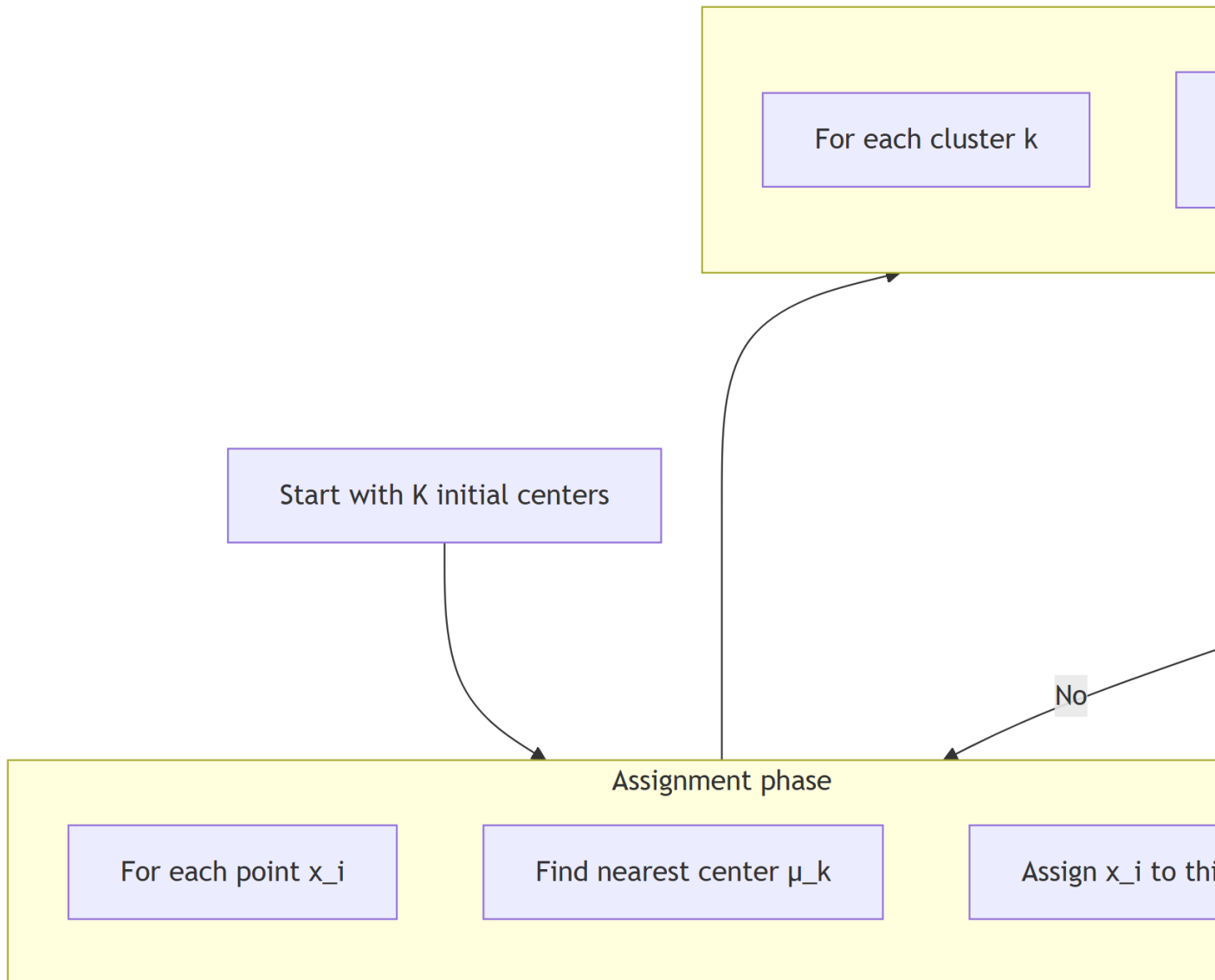
The Elected Representatives Analogy

Imagine a region with several villages. To reduce travel times, we want to build K administrative centers. How to place them optimally?

1. **Initialization:** Place K centers randomly

2. **Assignment:** Each village chooses the nearest center
3. **Readjustment:** Each center moves to the centroid of the villages that chose it
4. **Iteration:** Repeat until stable

This is the essence of **k-means**!



The Mathematical Heart: The Objective Function

Precise Formulation

Let:

- **Data:** $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N$, $\mathbf{x}_i \in \mathbb{R}^D$
- **Number of clusters:** K (fixed a priori)
- **Cluster centers:** $\mathbf{M} = \{\mu_k\}_{k=1}^K$, $\mu_k \in \mathbb{R}^D$
- **Assignment variables:** $\mathbf{Z} = \{z_{ik}\}$, where $z_{ik} \in \{0, 1\}$ and $\sum_{k=1}^K z_{ik} = 1$

The Function to Minimize

k-means minimizes the **sum of squared intra-cluster distances**:

$$J(\mathbf{M}, \mathbf{Z}) = \sum_{i=1}^N \sum_{k=1}^K z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

Where $\|\cdot\|_2$ is the Euclidean norm: $\|\mathbf{x}\|_2 = \sqrt{\sum_{d=1}^D x_d^2}$

Notation Details

Term	Meaning	Dimension
z_{ik}	Binary variable: 1 if $\mathbf{x}_i$ cluster k , 0 otherwise	Scalar
$\mathbf{x}_i$	Data point i	$\mathbb{R}^D$
μ_k	Center of cluster k	$\mathbb{R}^D$
$\ \mathbf{x}_i - \mu_k\ _2^2$	Squared Euclidean distance	Scalar

Crucial observation: For a given point $\mathbf{x}_i$, only one z_{ik} equals 1 (the one of the cluster it belongs to), the others equal 0. So for each point, we only count its distance to its own cluster center.

Optimization by Coordinate Descent

The Combinatorial Problem

Minimizing J simultaneously over $\mathbf{M}$ and $\mathbf{Z}$ is **NP-hard**: there are K^N possible assignments!

Strategy: Alternate Optimization

We use **coordinate descent**:

1. **Fix** $\mathbf{M}$, optimize over $\mathbf{Z}$ (E-step)
2. **Fix** $\mathbf{Z}$, optimize over $\mathbf{M}$ (M-step)
3. **Repeat** until convergence

E-step: Assignment Optimization

With $\mathbf{M}$ fixed, the problem becomes:

For each $i = 1, \dots, N$, choose z_{ik} to minimize:

$$\sum_{k=1}^K z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

Under constraint: $\sum_{k=1}^K z_{ik} = 1, z_{ik} \in \{0, 1\}$

Solution: Assign $\mathbf{x}_i$ to the cluster whose center is closest!

Formally:

$$z_{ik} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_i - \mu_j\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

M-step: Center Optimization

With $\mathbf{Z}$ fixed, the problem becomes:

For each $k = 1, \dots, K$, find μ_k minimizing:

$$\sum_{i=1}^N z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

Gradient Calculation

$$\nabla_{\mu_k} \left(\sum_{i=1}^N z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2 \right) = -2 \sum_{i=1}^N z_{ik} (\mathbf{x}_i - \mu_k)$$

Optimality Condition

Setting the gradient to zero:

$$\sum_{i=1}^N z_{ik} (\mathbf{x}_i - \mu_k) = 0$$

$$\Rightarrow \mu_k = \frac{\sum_{i=1}^N z_{ik} \mathbf{x}_i}{\sum_{i=1}^N z_{ik}}$$

Interpretation: The optimal center is the **mean** (center of mass) of the points in the cluster.

Proof of Convergence

Monotonic Descent

Let $J^{(t)} = J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)})$

1. **E-step:** We choose $\mathbf{Z}^{(t+1)}$ optimal for fixed $\mathbf{M}^{(t)}$:

$$J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)})$$

(strict inequality if at least one assignment changes)

2. **M-step:** We choose $\mathbf{M}^{(t+1)}$ optimal for fixed $\mathbf{Z}^{(t+1)}$:

$$J(\mathbf{M}^{(t+1)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)})$$

(strict inequality if at least one center changes)

3. **Transitivity:**

$$J^{(t+1)} = J(\mathbf{M}^{(t+1)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)}) = J^{(t)}$$

Consequence

The sequence $\{J^{(t)}\}_{t=0}^{\infty}$ is **decreasing** and **bounded below** by 0, therefore it converges.

Warning: It converges to a **local minimum**, not necessarily global!

Deep Relationship with Gaussian Models

From the EM Algorithm to k-means

To understand the connection, let's start from a **Gaussian Mixture Model** (GMM).

Gaussian Mixture Model

We assume data is generated by K Gaussians:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$$

Where:

- π_k is the weight of cluster k ($\pi_k \geq 0$, $\sum_k \pi_k = 1$)
- $\mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$ is a multivariate Gaussian

Latent Variables

Let's introduce latent variables $\mathbf{z}_i = [z_{i1}, \dots, z_{iK}]$ with $z_{ik} \in \{0, 1\}$ and $\sum_k z_{ik} = 1$, indicating which cluster $\mathbf{x}_i$ belongs to.

Complete Model

The joint distribution is written:

$$p(\mathbf{x}_i, \mathbf{z}_i) = \prod_{k=1}^K [\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)]^{z_{ik}}$$

The EM Algorithm for GMM

EM maximizes the log-likelihood of observed data with respect to parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.

E-step (Expectation)

We calculate the responsibility (posterior):

$$\gamma(z_{ik}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)}$$

This is a **soft assignment**: $\gamma(z_{ik}) \in [0, 1]$, $\sum_k \gamma(z_{ik}) = 1$.

M-step (Maximization)

We maximize the expectation of the complete log-likelihood:

$$Q(\theta, \theta^{\text{old}}) = \sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) [\log \pi_k + \log \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)]$$

The updates are:

$$\begin{aligned}\mu_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik}) \mathbf{x}_i}{\sum_{i=1}^N \gamma(z_{ik})} \\ \Sigma_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik}) (\mathbf{x}_i - \mu_k^{\text{new}})(\mathbf{x}_i - \mu_k^{\text{new}})^\top}{\sum_{i=1}^N \gamma(z_{ik})} \\ \pi_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik})}{N}\end{aligned}$$

Transition to k-means: Simplifying Assumptions

Assumption 1: Spherical and Equal Covariances

Assume $\Sigma_k = \sigma^2 I$ for all k , with σ^2 fixed and identical for all clusters.

The Gaussian density becomes:

$$\mathcal{N}(\mathbf{x}_i | \mu_k, \sigma^2 I) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{x}_i - \mu_k\|^2}{2\sigma^2}\right)$$

Assumption 2: Zero Variance Limit

Consider the limit when $\sigma^2 \rightarrow 0$ (very “peaked” distributions).

In this case, $\exp\left(-\frac{\|\mathbf{x}_i - \mu_k\|^2}{2\sigma^2}\right)$ decreases very rapidly with distance.

Assumption 3: Equal Weights

Assume $\pi_k = 1/K$ for all k (equiprobable clusters a priori).

Derivation of the k-means Objective Function

E-step in Hard Limit

When $\sigma^2 \rightarrow 0$, the responsibility $\gamma(z_{ik})$ becomes:

$$\gamma(z_{ik}) \rightarrow \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2 \\ 0 & \text{otherwise} \end{cases}$$

This is exactly a **hard assignment** like in k-means!

Simplified M-step

With these assumptions, the expectation Q becomes (up to an additive constant):

$$Q(\theta, \theta^{\text{old}}) \approx -\frac{1}{2\sigma^2} \sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) \|\mathbf{x}_i - \mu_k\|^2 + \text{constant}$$

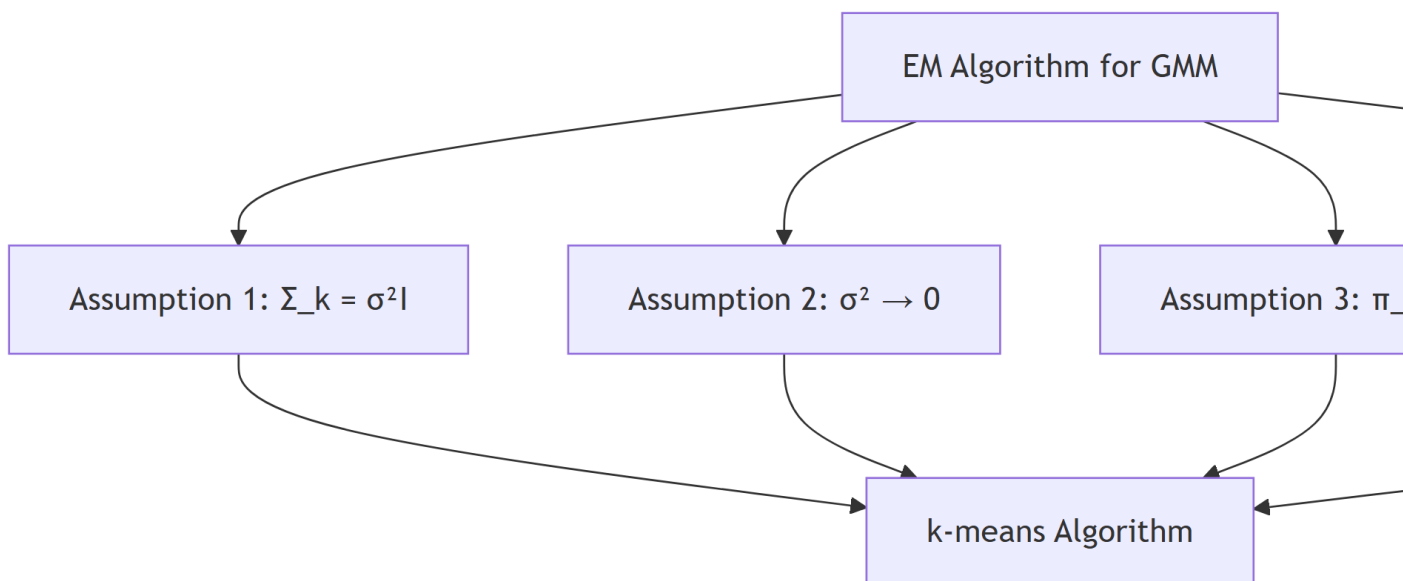
Maximizing Q is then equivalent to **minimizing**:

$$\sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) \|\mathbf{x}_i - \mu_k\|^2$$

With the hard assignment $\gamma(z_{ik}) = z_{ik}$, we recover exactly the function J of k-means!

Summary Table: EM $\rightarrow$ k-means

Aspect	General EM for GMM	k-means (limit case)
Assignment	Soft (probabilistic)	Hard (deterministic)
Covariances	Arbitrary Σ_k	$\Sigma_k = \sigma^2 I$
Variance	Learned from data	$\sigma^2 \rightarrow 0$
Weights	Learned π_k	$\pi_k = 1/K$
Objective function	Log-likelihood	Sum of squared distances



Practical Implications of This Connection

What This Teaches Us About k-means

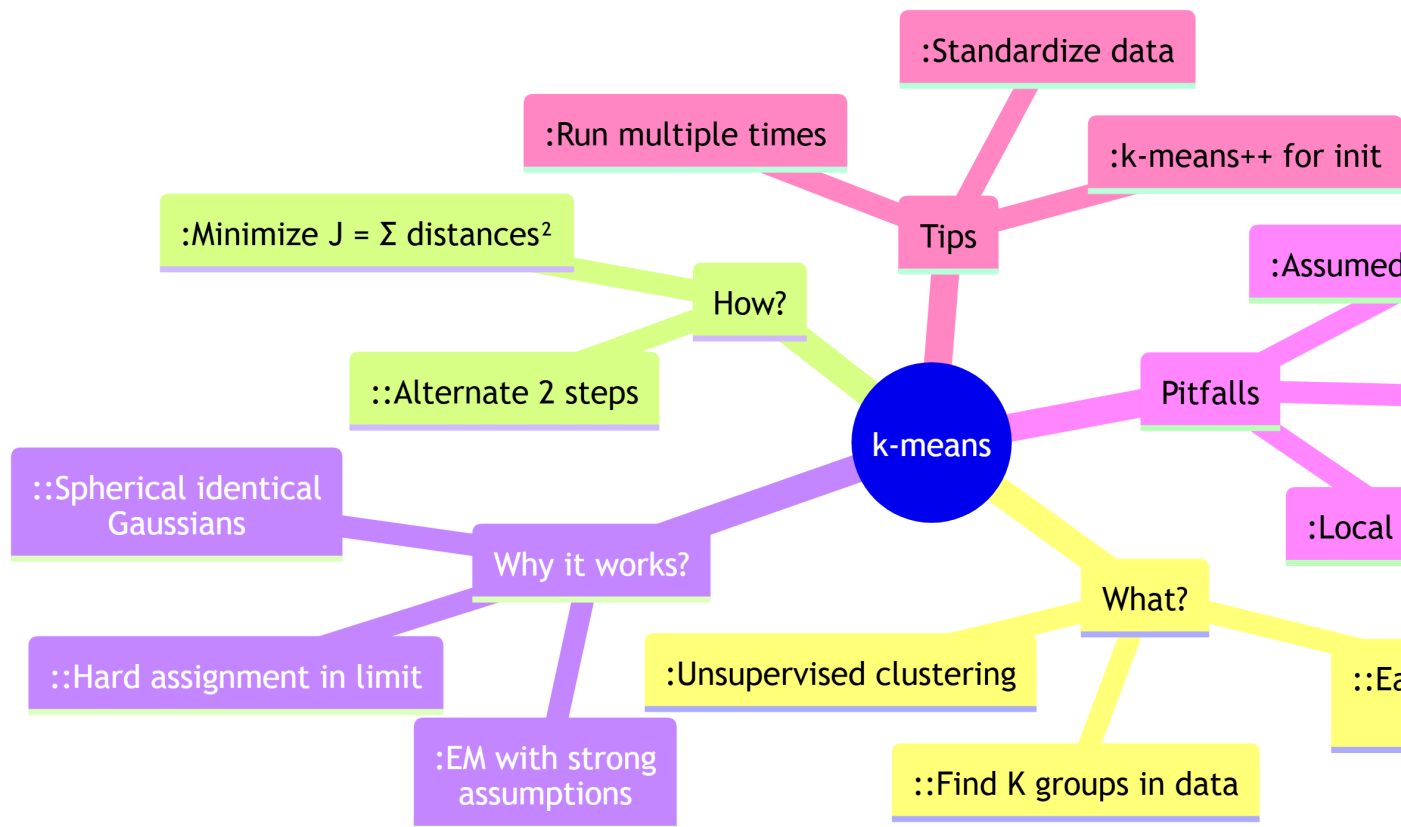
1. **Sphericity assumption:** k-means assumes **spherical** clusters (same variance in all directions)
2. **Scale sensitivity:** Since Gaussians have the same covariance, k-means is sensitive to data normalization
3. **Equal variance assumption:** All clusters are assumed to have the same “spread”
4. **Low overlap limit:** k-means works better when clusters are well separated

Consequences for Application

- **Normalization necessary:** Always normalize data before applying k-means
 - **Cluster shape:** k-means detects elongated or complex-shaped clusters poorly
 - **Choice of K:** More difficult when clusters have very different variances
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Final Intuitive Version

The Essence in One Image



Key Takeaway

k-means is a special case of EM where we assume each cluster is a spherical Gaussian with the same variance, and we take the limit where these Gaussians are infinitely “peaked” (low variance).

This theoretical understanding explains:

- **Why** k-means works so well on certain problems
- **When** to use it (compact, spherical clusters)
- **How** to improve it (good initialization, normalization)

The genius of k-means is having **simplified to the maximum** a complex problem to obtain a fast, simple, and effective algorithm in many practical cases.

For the More Curious: Further Reading

Possible Generalizations

1. **k-medoids**: Uses an actual point as center (less sensitive to outliers)
2. **Fuzzy C-means**: Probabilistic assignment like EM
3. **k-means with different metrics**: Manhattan distance, cosine...

Open Challenges

- **Automatic choice of K**: Elbow method, information criteria...
- **Clusters with variable densities**: DBSCAN as alternative
- **High dimensions**: Curse of dimensionality

The beauty of k-means lies in its balance between **mathematical simplicity**, **computational efficiency**, and **intuitive interpretability** - a rare and powerful combination in data science.